WEEK 01 : TRIAL PRACTICE PHASE : DAYS 01 TO 05

Foundations, Data Handling & Exploratory Analysis

Development Environment Scaffold, Leak-Free Ingestion & Statistical EDA Practice

PROGRAM & ROLE	CURRICULUM PHASE	TIME COMMITMENT	SCHEDULE & FEEDBACK
ML Engineering Intern (All-Round)	**Trial Practice Tasks (Weeks 1-2)**	**15-20 Hours/Week (Part-Time Unpaid)**	**Submit by Friday (11:59 PM) | Weekend Feedback**

01 Program Overview & Trial Practice Strategy

PRACTICE PHASE

Welcome to the Attenomy AI & Machine Learning Engineering team. This 1-month intensive program is divided into two distinct halves: (1) **Trial Practice Phase (Weeks 1-2)**, where you complete structured trial tasks on real-world benchmark datasets to build rock-solid engineering habits, leak-free pipeline discipline, and model evaluation rigor; and (2) **Product Contribution Phase (Weeks 3-4)**, where you directly contribute to our enterprise AI product: **Attenomy Cortex** (our autonomous multi-agent orchestration and cognitive knowledge retrieval platform).

02 Repository Scaffold & Environment Setup

DAYS 1-2

```
# Initialize personal practice repository and scaffold directory structure
$ mkdir attenomy-ml-practice && cd attenomy-ml-practice && git init
$ python -m venv .venv && source .venv/bin/activate # Windows: .\.venv\Scripts\Activate.ps1
$ pip install --upgrade pip && pip install pandas numpy scikit-learn matplotlib seaborn pytest ruff
pyarrow
$ pip freeze > requirements.txt && mkdir -p data/raw data/clean notebooks src tests reports/figures
models
$ git add . && git commit -m "feat: initial repository scaffold"
```

TASK 1.1 Personal Git Repository & Ingestion Utility Engineering

Days 1-2 : Mon-Tue

Establish professional commit hygiene (minimum 8 atomic commits over 2 days) and construct 3 core data handling functions in `src/utils.py` backed by a comprehensive `pytest` test suite in `tests/test_utils.py`.

- **Function 1: load_data(path: str) -> pd.DataFrame** : Ingests JSON/CSV tabular datasets, verifies required schema columns, performs defensive type casting, and raises descriptive exceptions on missing files.
- **Function 2: summarize_dataframe_health(df: pd.DataFrame) -> dict** : Computes row/column dimensions, column data types, null percentage per feature, duplicate row counts, and memory footprint in megabytes.
- **Function 3: train_val_test_split(df: pd.DataFrame, ratios: tuple, seed: int, stratify_col: str = None) -> Tuple[pd.DataFrame, pd.DataFrame, pd.DataFrame]** : Partitions data into clean train, validation, and test splits with fixed random seeds.
- **Unit Tests:** Implement at least 3 distinct tests in `tests/test_utils.py` testing empty file handling, schema shape consistency, and split ratio accuracy.

ACCEPTANCE CRITERIA

✓ Fresh clone executes **pip install -r requirements.txt && pytest** with zero test failures.
✓ All functions in **src/utils.py** possess complete Google-style docstrings with explicit type hints and doctests.
✓ Codebase adheres to strict linting: **ruff check .** completes with zero warnings or errors.

TASK 1.2 Trial Data Cleaning & Schema Normalization

Days 2-3 : Tue-Wed

Ingest a complex real-world tabular practice dataset (such as enterprise service interaction logs, churn records, or user engagement telemetry with 20,000+ entries) containing missing categories, outliers, and corrupted fields. Build a reproducible cleaning pipeline.

- **Missing Value Audit:** Identify null entries across numeric and categorical features. Explicitly document imputation strategies: median for skewed numeric variables, domain-specific constants ("UNKNOWN") for categorical features, and drop corrupted primary keys.
- **Anomaly & Outlier Mitigation:** Detect invalid values, inconsistent casing, duplicate entries, and extreme numeric outliers using interquartile range (IQR) or z-score thresholds.
- **Encoding & Scaling Preparation:** Specify one-hot encodings for nominal categories, ordinal mappings for ordered values, and standard scalers for continuous attributes.
- **Data Dictionary:** Author **`reports/data_dictionary.md`** documenting 100% of dataset columns with data types, business definitions, allowable ranges, and missing value policies.
- **Script Packaging:** Export reproducible cleaning logic to a standalone Python script: `python src/cleaning.py --input data/raw/dataset.csv --output data/clean/clean.parquet`.

ACCEPTANCE CRITERIA

✓ Notebook **notebooks/01_cleaning.ipynb** runs top-to-bottom without warnings via "Restart & Run All".
✓ Zero unexplained missing values remain; output saved as high-efficiency **data/clean/clean.parquet**.
✓ Data dictionary covers 100% of schema attributes with complete data governance annotations.

TASK 1.3 Exploratory Data Analysis & Executive Insights Memo

Days 4-5 : Thu-Fri

Conduct rigorous statistical exploratory data analysis in **`notebooks/02_eda.ipynb`**, test formal hypotheses regarding data distributions, and author a high-impact 2-page executive insights memo for technical leadership.

- **8 Mandatory Visualizations:** Generate and export 8 publication-grade charts to `reports/figures/`: (1) Feature distributions, (2) Target variable balance, (3) Correlation heatmap, (4-6) Three feature-versus-target plots, (7) Outlier boxplots across subgroups, and (8) Temporal or interaction heatmap.
- **Hypothesis Formulation & Testing:** State at least 5 empirical hypotheses regarding data relationships. Formally test at least 2 using statistical methods (e.g. Chi-Square test of independence, two-sample t-test, or Mann-Whitney U test). Report test statistics and p-values.
- **Executive Insights Memo:** Author a 2-page document (**`reports/week1_insights_memo.pdf`**) synthesizing data findings, identifying operational risks (target imbalance, label leakage, collinearity), and defining the Week 2 classical modeling plan.

ACCEPTANCE CRITERIA

✓ Every figure has explicit descriptive titles, axis units, labeled legends, and a one-sentence takeaway caption.
✓ Statistical tests report null hypothesis, test statistic, degrees of freedom, p-value, and business interpretation.
✓ Executive memo delivers clear technical findings without unexplained jargon, directly reviewed by Attenomy ML Lead.

ATTENOMY ENGINEERING STANDARD : DATA LEAKAGE PREVENTION

Never compute preprocessing statistics across the complete dataset prior to splitting. Any scaler mean, imputer median, or encoder mapping derived from validation or test splits represents data leakage. Preprocessing transformations must always be fit strictly on training splits.

03 Week 1 Deliverables & Artifacts Matrix

VERIFICATION MATRIX

ARTIFACT NAME	REPOSITORY TARGET PATH	PRIMARY REQUIREMENT	EVALUATION
Utility Module	`src/utils.py`	3 typed utilities with docstrings & doctests	Pass / Fail
Unit Test Suite	`tests/test_utils.py`	Minimum 3 pytest tests passing with 0 errors	Pass / Fail
Cleaning Notebook	`notebooks/01_cleaning.ipynb`	Deterministic execution, zero null residuals	Score: 12%
Data Dictionary	`reports/data_dictionary.md`	100% schema documentation with null policies	Score: 12%
Clean Dataset	`data/clean/clean.parquet`	Snappy-compressed Parquet with verified dtypes	Pass / Fail
EDA Notebook	`notebooks/02_eda.ipynb`	8 figures, 5 hypotheses, 2 formal stat tests	Score: 12%
Executive Insights Memo	`reports/week1_insights_memo.pdf`	2-page memo: operational risks & ML targets	Score: 12%

04 Weekly Submission Deadline & Weekend 1-on-1 Feedback

GATE 1 ASSESSMENT

Weekly Submission Deadline: There is no fixed daily stand-up. Interns work flexibly (15-20 hours/week) and must submit all deliverables, notebooks, code commits, and reports to their personal practice repository by end of Friday (11:59 PM). On Saturday or Sunday, you will be invited to an online 1-on-1 meeting for personalized feedback, code review, and mentorship guidance. Passing Gate 1 unlocks Week 2 classical model training.

GATE 1 PASSING CONDITION & MENTOR PROBE QUESTIONS

✓ **Mandatory Pass Condition:** Repository cleanly structured, all pytest unit tests passing, clean data reproducible via script, and 2-page insights memo approved.

- **Technical Probe 1:** "Show me how your split logic ensures no data leakage. What steps ensure categorical encodings only reference training rows?"
- **Technical Probe 2:** "Walk through the statistical assumptions of the hypothesis test you conducted. How did you handle skewed variables or outliers?"
- **Technical Probe 3:** "Which columns exhibited the highest correlation with the prediction target, and how do you plan to handle potential collinearity in Week 2?"

05 Actionable Daily Execution Checklist

INTERN CHECKLIST

- ☐ **Day 1:** Clone repo, configure virtualenv, verify requirements.txt, push Day-0 commit.
- ☐ **Day 2:** Implement 3 utility functions in src/utils.py, write tests, verify pytest passes.
- ☐ **Day 3:** Profile raw dataset, execute cleaning logic in notebook, export Parquet file.
- ☐ **Day 4:** Complete data dictionary, plot 8 EDA figures, run statistical hypothesis tests.
- ☐ **Day 5:** Draft 2-page insights memo, push code, submit by end of Friday (11:59 PM), attend weekend 1-on-1 review.
- ☐ **Reflect:** Log 3 technical entries in LEARNING_LOG.md with concepts and code links.

06 Flexible Engineering Cadence & Mentorship Support SLA

ENGINEERING CADENCE

Flexible Workflow: No fixed daily stand-ups. Interns manage their hours flexibly to meet the 15-20 hours/week commitment while pushing regular atomic git commits (`feat:`, `fix:`, `test:`).

Weekend Feedback: Every Saturday or Sunday, you will receive personalized 1-on-1 feedback and code review from an Attenomy ML Lead.

45-Minute Help Escalation Rule: When blocked by a bug, investigate independently for 45-60 minutes before messaging your mentor with: (1) your objective, (2) the error traceback, and (3) your troubleshooting attempts.